DESCRIPTIVE STATISTICS #2

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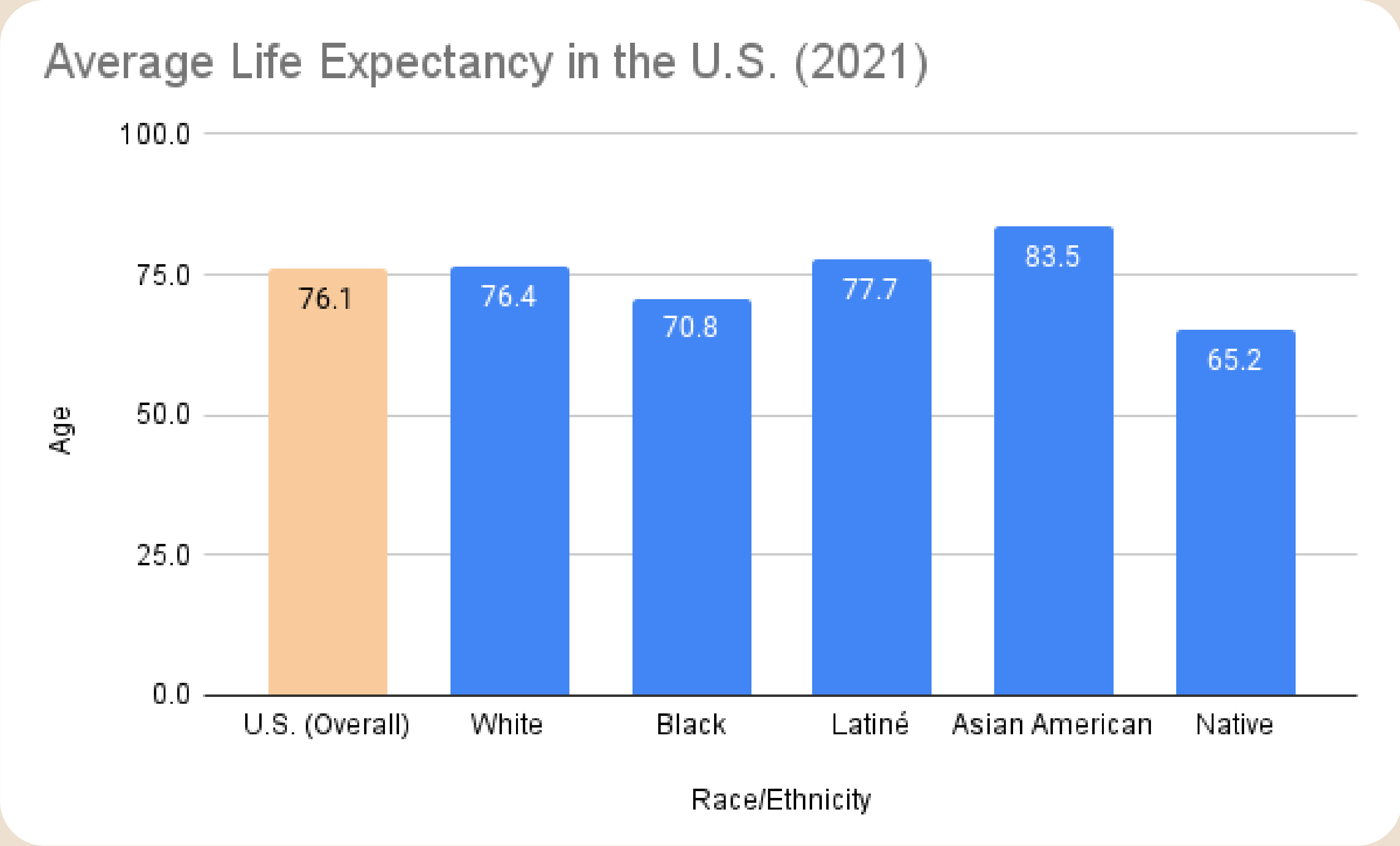
TOPICS

- Measures of central tendency
- Measures of dispersion/variability

MEASURES OF CENTRAL TENDENCY

- Give us an idea about the **average** or **typical** case in the distribution
- A **single value** that can describe a variable -- identifying a **central position of that variable** within a data set
- Powerful in that they reduce huge arrays of data to a single number



MODE: MOST FREQUENT VALUE

- Found by taking your values, arranging them chronologically, then **selecting the value that shows up most often**
- More than one mode is possible

- **Example 1:**
  - Scores: 92, 88, 74, 92, 53, 74, 87, 92, 74, 75, 55, 92, 88
  - Arranged: 53, 55, **74**, **74**, **74**, 75, 87, **88**, **88**, **92**, **92**, **92**, **92**

- **Example 2:**
  - We can also calculate the mode with data measured at the nominal level

Religion	Frequency
Protestant	128
Catholic	57
Evangelical	10
None	32
Other	15
Total	242

MEDIAN: THE MIDDLEMOST SCORE

- Always represents the **exact center of a distribution**
  - Half of the numbers are smaller and the other half are greater than the median

- **Example 1: Odd number of cases**
  - Scores: 92, 88, 74, 92, 53, 74, 87, 92, 74, 75, 55, 92, 88
  - Arranged: 53, 55, 74, 74, 74, 75, **87**, 88, 88, 92, 92, 92, 92
- **Example 2: Even number of cases**
  - Average of the two middle cases
  - Arranged: 53, 55, 74, 74, 74, 75, **87**, **88**, 88, 92, 92, 92, 92, 97

(87+88)/2 = **87.5**

MEAN: THE AVERAGE VALUE

- The arithmetic average is the **most commonly used** measure of central tendency
- To compute: Add all scores an then divide by the total number of scores (N)

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots\ldots\ldots\ldots + x_n}{n}$$

MEAN IS SUSCEPTIBLE TO EXTREMES

- When a distribution has a few extremely high or extremely low scores, the mean may be misleading as a measure of central tendency

- The **mean** and **median** will be <u>the same only when the distribution is symmetric</u>
  - When a distribution has some <u>extremely high scores</u> (**positive skew**), the mean will have a greater numerical value than the median
  - When a distribution has some <u>extremely low scores</u> (**negative skew**), the mean will have a lower numerical value than the median

- **Example:**

Group 1	Group 2
Values: 15, 20, 25, 30, 35	**Values**: 15, 20, 25, 30, 3500
Mean: 125/5 = 25	**Mean**: 3590/5 = 718
Median: 25	**Median**: 25

SYMMETRIC & SKEWED DISTRIBUTIONS

The skewness of a distribution refers to how far the distribution **deviates** from symmetry (a normal distribution)

- A distribution can be **positively-skewed** (AKA right-skewed) or **negatively-skewed** (AKA left-skewed)
- Skewness refers to the position of the "longer tail" of the distribution. For example, if the tail is on the left, it is a negatively-skewed (AKA left-skewed) distribution.

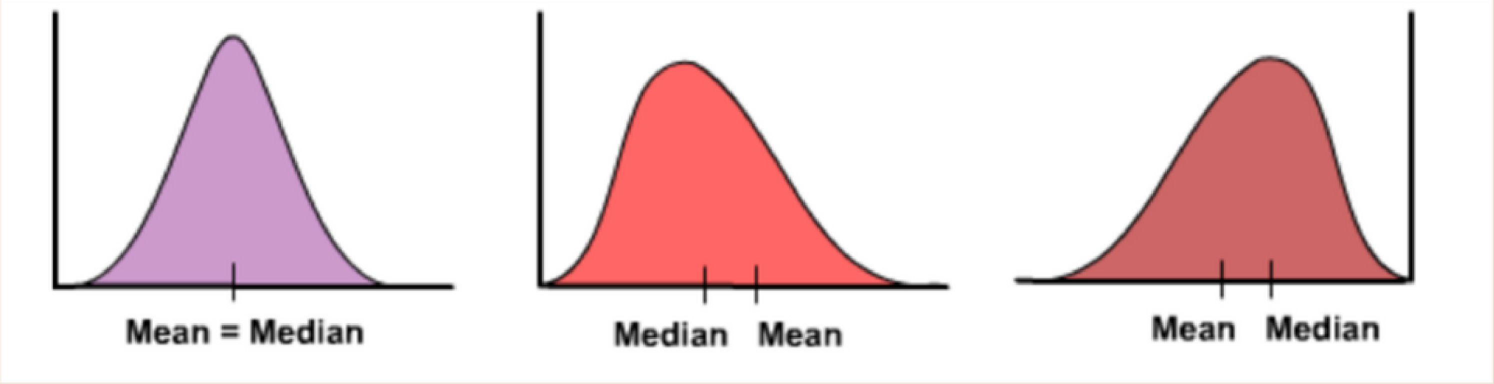
A classic example of a positively-skewed (AKA right-skewed) distribution is *income or salary*, where high-earners provide a false representation of the typical income if expressed as a mean and not a median

An example of a negatively-skewed (AKA left-skewed) distribution is *age of retirement*, where few people retire younger, and most people retire at older ages.

Symmetrical
Normal Distribution

Positive Skewness
Right Skew

Negative Skewness
Left Skew



MEASURES OF DISPERSION/VARIABILITY

- Whereas measures of central tendency allow us to locate the average or typical scores, measures of dispersion provide information about the amount of **spread** or **clustering** amongst the scores
  - Common: standard deviation, variance, and range

- A good measure of dispersion should:
  - Describe the average deviation of the scores
    - Describe **how far**, on average, the scores are from the center of the distribution
  - Decrease in value as more scores are included

DISPERSION: BEYOND MEANS

- Suppose a manager wanted to evaluate the **performance of two delivery truck drivers**. They look at each driver's mean time per delivery (sum of minutes spent for each delivery / total # deliveries), and find that the means of the two drivers are *<u>very close</u>* to each other.
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- Measures of dispersion are important, because reveal additional information -- about the **spread** (of delivery times) around the mean:
  - Driver A:
    - Scores are more grouped around the mean
    - More consistent/less spread of values
  - Driver B:
    - More scores in the high (long delivery times) and low (short delivery times) ranges.
    - More diverse/variety

STANDARD DEVIATION: THE AVERAGE DEVIATION

Standard deviation is a widely-used measure of dispersion

- Is the **average deviation from the mean**
  - Deviation (AKA raw score deviation) is the distance between each value and the mean
  - If values are clustered around the mean, deviations will be small

- Similar to calculating an average, we **sum the deviations**, and divide by the number of cases...
  - But *the sum of the deviations is always zero*, so to deal with this (and get rid of negatives), we *square* each of the deviations

- Example:
  - Five scores: 10, 20, 30, 40, 50
  - **Mean** = 150/5 = 30

	Values	Deviations	Squared Deviations
	10	(10-30) = -20	(-20)^2 = 400
	20	(20-30) = -10	(-10)^2 = 100
	30	(30-30) = 0	(0)^2 = 0
	40	(40-30) = 10	(10)^2 = 100
	50	(50-30) = 20	(20)^2 = 400
Sum	150	0	1000

The sum of squared deviations will increase with larger samples

STANDARD DEVIATION CONTINUED

- Taking the average deviation (or, rather, the **average of the sum of squared deviations**) is a way to standardize for samples of different sizes
  - Divide the sum of squared deviations by sample size – or rather, **an "adjusted" sample size**. So rather than dividing by N, we divide by "N-1"

  - Doing so, gives us one measure of dispersion, called **Variance**:

$$Variance = \frac{\Sigma(x - \bar{x})^2}{(n-1)}$$

  - But, remember, we squared the deviations so we could get rid of the negatives (and the sums resulting in 0).

  - To put these values back into their original scale, we take the square root of variance, which gives us the **Standard Deviation**:

$$SD = \sqrt{Variance} = \sqrt{\frac{\Sigma(x - \bar{x})^2}{(n-1)}}$$

STANDARD DEVIATION CONTINUED

Standard deviation will be **higher** for the more diverse distributions and **lower** for less diverse distributions

When standard deviation is 0, every single case in the sample has exactly the same score

Typically, for measures of dispersion/variability, we use

- **range** – which tells us about the extremes of the distribution, like the lowest and highest values, and
- **standard deviation** – which tells us about how close or clustered the values are around the mean.
  - Variance can simply be thought of as "*a stepping stone towards calculating the standard deviation*"